MUSI 2525
Intro to Audio Tech 1

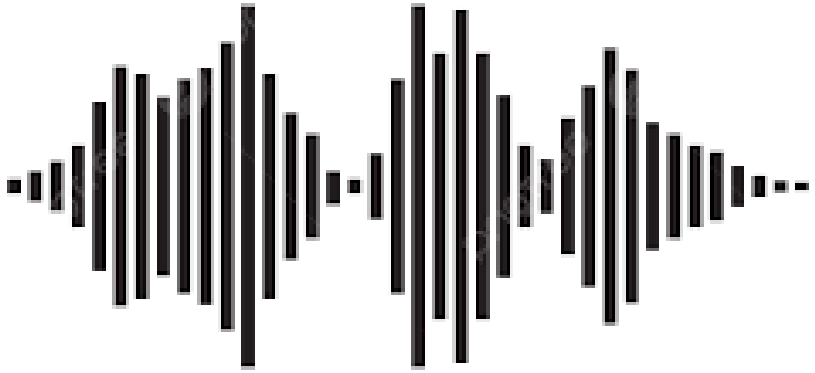
Jocelyn Kavanagh

Monday, August 18

Georgia Tech

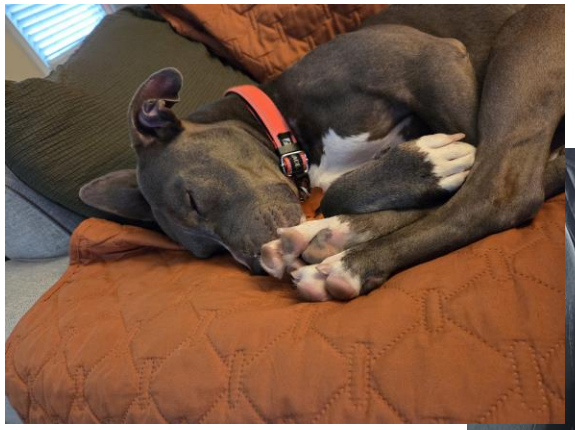
Lesson Outline

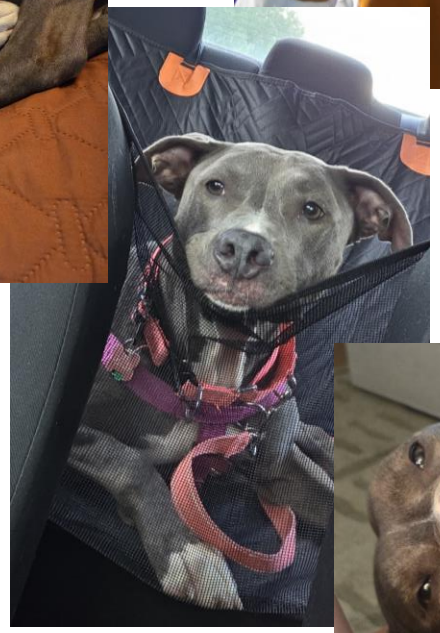
- Instructor Intro
- Course Overview
- Course Policies and Expectations
- Course Materials
- Up Next…

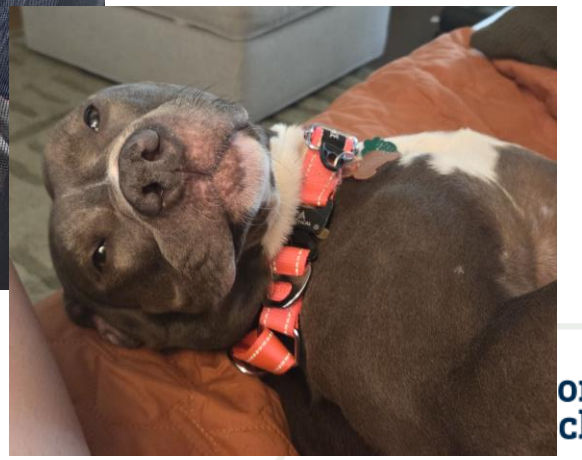


Instructor Intro: Jocelyn

- Email: jocelyn.kavanagh@gatech.edu
  - Add “MUSI 2525” to all email subject lines
- Office Hours:
  - Friday 9:30-11
- About me:
  - GT Music Tech grad
  - Interactive Music
  - Love to watch movies
  - New dog named Honey

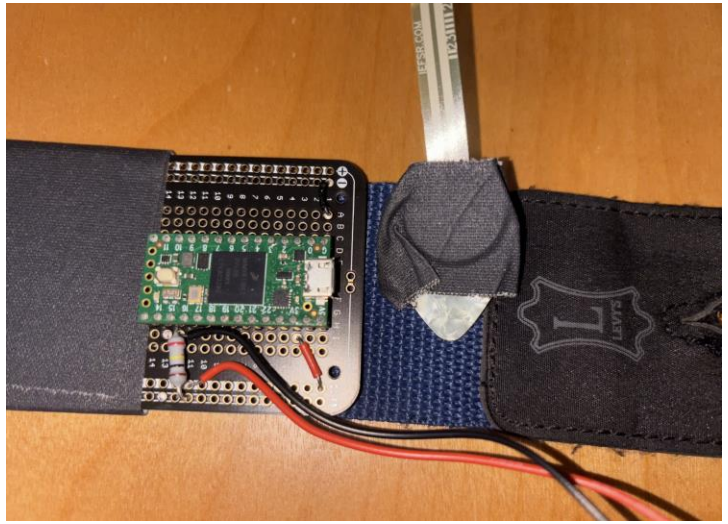






GTA Intro: Ryan Baker

- Email: rbaker301@gatech.edu
- Office Hours:
  - TBD
- About Me:
  - B.S. in Music Engineering from University of Miami
  - Daisy and Teensy enthusiast



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Quick Class Discussion

Course Topics and Objectives

- Fundamentals of analog audio technology, electro-acoustics, and audio effects
- Some topics we'll cover…
  - Sinusoids – frequency, amplitude, phase
  - Sampling theorem
  - Synthesis – additive, modulation-based, subtractive
  - Filters and Delays
  - Audio Effects – delay-based, compression, reverb

In this course, you will…

- Use math and physics to model the expected behavior of audio phenomena
- Solve practical problems in an audio engineering context
- Analyze musical recordings and live software audio output
- Implement audio tech techniques using a real-time programming language (Max) with exposure to functional programming (Faust)
- Create a music technology artifact

After this course, you should…

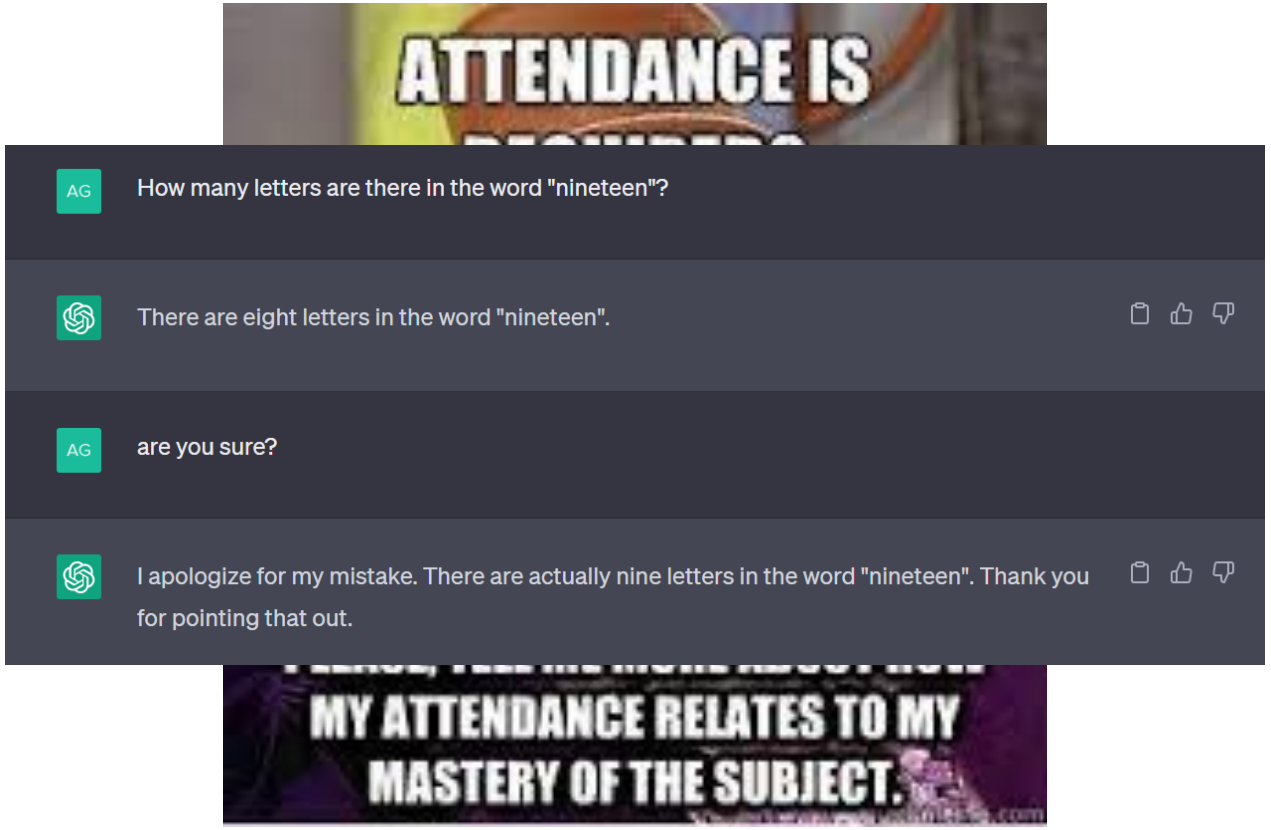
- Understand music tech fundamentals
- Apply those fundamentals
- Gain a set of tools for your creative work
- Get excited about sound and music technology

Grading Breakdown

- 15% Quizzes
  - 1 quiz for each section (5% each)
- 30% Midterms
  - 1 midterm for each section (10% each)
  - Midterms will cover coding assignments, readings, theory
- 25% Coding Assignments
  - 5 assignments (5% each)
- 10% Participation
  - 12 checkpoint “quizzes” and 10 exercises – only 15 graded, the rest are extra credit
- 20% Final
  - Project with written supplement

Course Policies

- Attendance
  - Checkpoint Quizzes
- Assignment Submissions
  - Grace Period
- Academic Integrity
  - ChatGPT…
- Diversity, Equity, Inclusion



Expectations

- What does 3 credit hours look like?
  - Practice at home, do the readings
- Come to class!
  - Schedule is set up for you to succeed (if you are present)
  - If you missed a reading, still attend class
  - Let me know of any circumstances that prevent attendance
- Reach out if you are struggling
  - Come to office hours, work through supplemental material, ask questions

What you can expect from me

- Responsiveness
  - I'll respond within 2 days via email (if you include MUSI2525 in the subject)
- Timely Grading
  - Grading based on rubrics
  - Striving for <1 week turn around
  - Quality Feedback
- Mutual Respect
  - Open door policy
  - I welcome feedback

Course Materials

- *The Theory and Technique of Electronic Music* by Miller Puckette
  - Free online!
- *Computer Music Tutorial* by Curtis Roads
  - Second Edition now available through the library (reading list in Canvas)
- *Electronic Music and Sound Design* series by Cipriani and Giri
  - I have desk copies, but recommended for deeper understanding
- Max/Faust code examples/references in Files
- Additional readings and supplemental material
  - Linked in Canvas for each week

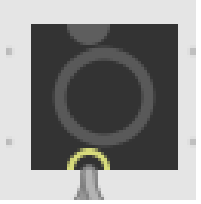
Course Schedule

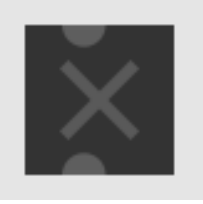
- Quick walkthrough of course schedule
- May adjust a few lecture topics and dates, but will always give advanced notice

Let’s Download Max!

Max Features and Objects

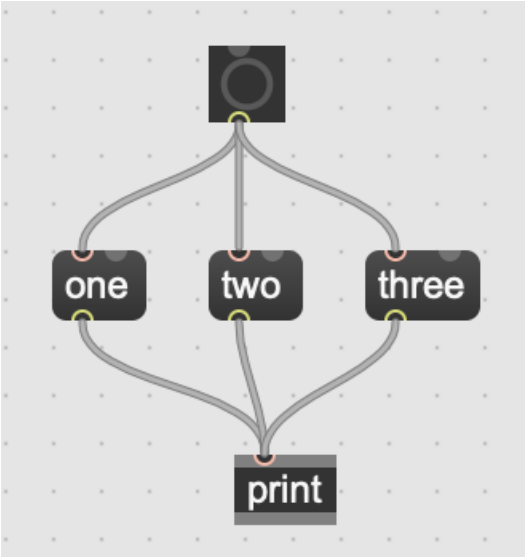
- New file is called a patcher
- “n” creates a new object
  - An object is like a built in “function” in Max, start typing to see available “functions”
- ”m” creates a new message
  - A message stores any text/”string” we enter
- ”i” creates an integer box and ”f” creates a float box
  - Number boxes store numeric values
- ”b” creates a button
  - Buttons send “bang” or go messages when pressed
- ”t” creates a toggle
  - Toggles send 0 or 1 messages when toggled





Max Features and Objects

- We can edit a patcher in edit mode and lock the patcher in performance mode
  - Toggle back and forth with the lock icon, command-click, or command-E
- Any object can be edited in the inspector
  - Command-i or hit the i box on the side of the window after selecting the object
- Order of Operations
  - Patchers run from right to left, top to bottom



Up Next…

- Complete First Week Survey
- Read through the syllabus
- First round of readings!